

NGUYEN LE GIA HUY

Backend Developer Intern / Software Systems Engineer

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PROFESSIONAL SUMMARY

Software Engineering graduate specializing in **Backend Infrastructure, Distributed Systems & Concurrency Control**. Hands-on experience architecting high-performance REST APIs & Microservices with **NestJS, Express.js, and Python**. Experienced in resolving high-concurrency race conditions using **PostgreSQL row-level locking (SELECT FOR UPDATE)**, semantic question retrieval with **pgvector**, cache acceleration with **Redis**, sub-second event pushing with **Socket.IO**, and automated **ETL / RAG ingestion pipelines**. Comprehensive security with **JWT & OAuth 2.0, RBAC, and HMAC-SHA256 Idempotency**. **Available for full-time internship and employment.**

BACKEND SKILLS

Languages:	TypeScript, JavaScript, Python, SQL, C#, Dart, HTML5/CSS3
Backend & APIs:	NestJS Microservices, Express.js, Node.js, RESTful APIs, Socket.IO, BullMQ
Databases & Caching:	PostgreSQL (pgvector, pg_trgm), Prisma ORM, Redis, Neon Serverless, MySQL
AI, DevOps & Security:	Gemini API (RAG), Docker, Linux, Git, Playwright/Jest, JWT, OAuth 2.0, RBAC, HMAC-SHA256

FEATURED PROJECTS

09/2025 – 03/2026

TechStore (Vanfio) — High-Concurrency E-Commerce Platform & NestJS Backend

Role: **Backend / Database Developer** | Demo: website-for-aunt.vercel.app | Source: github.com/nlgiahuy01112003/techstore

NestJS · PostgreSQL · Prisma ORM · Redis · SELECT FOR UPDATE · MoMo Webhook · Socket.IO · Docker

Enterprise tech hardware e-commerce backend featuring atomic transaction blocks, row-level locking, and real-time chat routing.

- **Concurrency Control & Row-Level Locking:** Enforced PostgreSQL `SELECT FOR UPDATE` within Prisma database transactions, eradicating 100% of race conditions and preventing inventory overselling during simultaneous peak checkouts.
- **Idempotent Webhook Processing:** Verified MoMo and VNPay payment webhook callbacks using HMAC-SHA256 signatures, request deduplication keys, and a 10-minute automated voucher reservation lock.
- **Optimized Trigram Search (pg_trgm):** Integrated native PostgreSQL Full-Text Search with `pg_trgm` similarity for VI/EN/ZH synonyms, eliminating external search engine memory footprint while keeping query latency < 50ms.
- **Multi-Tier Redis Caching:** Designed Redis cache layers for user sessions and hot product catalogs, reducing direct database load by 45% and ensuring consistent API latency under load.
- **Defense-in-Depth Security:** Configured rotating JWT access tokens, TOTP multi-factor authentication for admins, strict DTO ValidationPipes, RBAC guards, and Helmet headers against DDoS/Brute-force.

03/2026 – Present

HyperGlot — Backend API, pgvector & Japanese Learning Ecosystem (Turborepo Monorepo)

Role: **Backend / System Architect** | Monorepo: github.com/nlgiahuy01112003/HyperGlot

NestJS · PostgreSQL · Prisma ORM · Neon pgvector · Redis · BullMQ · scrypt Auth · Playwright · Docker

JLPT N5-N1 Japanese learning platform featuring a NestJS API Gateway, Neon PostgreSQL Serverless, and semantic vector retrieval.

- **Monorepo Architecture & Type Contracts:** Architected `apps/api` (NestJS) and `packages/database` (Prisma) sharing end-to-end TypeScript contracts with web, admin, and mobile clients via Turborepo, eliminating contract drift.
- **Semantic Search with Native pgvector:** Avoided costly external vector database services by leveraging PostgreSQL's native `pgvector` extension on Neon Serverless for semantic JLPT question retrieval.
- **Self-Hosted Secure Auth (scrypt):** Built an in-house auth module using scrypt key derivation, Redis session storage, and token rotation, removing external vendor dependencies.
- **Automated Testing & Security Audits:** Implemented 60+ automated test cases with Playwright cross-browser E2E coverage for auth/lesson flows, achieving WCAG 2.1 AA accessibility and OWASP Top 10 compliance.

OTHER CORE PROJECTS

06/2026 – 07/2026

OptiBot RAG System (KB-Harvest) — Grounded AI Assistant & ETL Ingestion Pipeline

Role: **Python / Backend Engineer** | Source: github.com/nlgiahuy01112003/kb-harvest

Python | Google Gemini API | BeautifulSoup4 | SHA-256 Delta Sync | Docker | GitHub Actions Cron

- Automated ETL & RAG pipeline: Ingests Zendesk Help Center documentation via paginated REST APIs, stripping HTML into token-efficient Markdown (saving over 40% LLM tokens).
- **SHA-256 Delta Synchronization:** Implemented checksum hashing via `state.json` to detect newly added or modified articles, skipping unchanged files and slashing daily uploads by **~85%**.
- **Gemini File Search RAG:** Configured Gemini File Search (512-token chunks, 100-token overlap) to answer CLI engineering queries with exact source URL citations.
- **Containerization & CI/CD Cron:** Packaged as a single-command Docker container, scheduled via **GitHub Actions cron (02:00 UTC)** with automated artifact logging.

08/2026 – 09/2026

AI Mock Exam Platform — NestJS Backend & SFIA L3-4 AI Assessment Engine

Role: **Backend / Full-Stack Engineer** | Demo: ai-exam-live | Source: GitHub

NestJS | Next.js | Neon PostgreSQL (SSL) | Pyodide WebAssembly | Gemini AI Grader | Jest | REST APIs

- **Dual-Mode Database & AI Grader:** Built a NestJS REST API with Neon PostgreSQL Serverless (SSL) serving **1,020+ SFIA L3-4 questions**, featuring **Gemini AI Code Grader** and automatic **Offline Fallback**.
- **Comprehensive API Test Suite (100% Pass):** Authored **24+ automated API integration test cases** covering exam randomization, score computation across 1,020+ questions, and Smart Mentor AI endpoints.

03/2024 – Present

POMI Milk Tea — Express.js Backend, Recipe Inventory & Real-Time Ordering System

Role: **Backend Developer** | Source: github.com/nlgiahuy01112003/pomi_web

Express.js | PostgreSQL Pool | Socket.IO | Gemini AI & Zalo Alert | Goong.io API | Idempotency

- **Core REST API & Socket.IO Gateway:** Built Express TypeScript REST APIs and PostgreSQL pools, pushing live order payloads to kitchen terminals under 800ms with Idempotency Key **deduplication**.
- **Recipe Inventory & AI Alerts:** Engineered a per-cup Recipe & Ingredient Inventory engine (Base ingredients vs. Toppings), integrated a **Gemini AI Assistant**, and automated **Zalo low-stock alerts**.
- Integrated Goong.io APIs to compute branch routing and dynamically evaluate real-time transit distances.

EDUCATION

09/2021 – 03/2026

Engineer's Degree in Information Technology (Software Engineering)

HUTECH University — GPA: 2.86/4.0

- **Core foundation:** Data structures & algorithms, object-oriented programming (OOP), relational databases, operating systems, computer networks, software architecture, and automated testing.
- **Backend practice:** Database schema design, API contracts, atomic transactions, authentication, authorization, and CI/CD pipelines.

CERTIFICATIONS & LANGUAGES

- **CEFR B1 English — HUTECH:** General English Test at Intermediate Level; issued Dec 4, 2025.
- **Udemy — Flutter Development Bootcamp with Dart:** Dr. Angela Yu; Dart OOP, Flutter State Management & APIs.
- **Communication & Teamwork — HUTECH:** Issued Apr 16, 2025; technical problem solving and cross-team collaboration.
- **Udemy — Full-Stack Web Development Bootcamp:** Dr. Angela Yu; React, Node.js, Express, REST APIs.
- **Udemy — 100 Days of Code™ Python Pro Bootcamp:** Dr. Angela Yu; Python OOP, Web Scraping, Automation.

PROFESSIONAL EXPERIENCE

12/2025 – 06/2026

IT Helpdesk / Manual Tester & System Deployment

EMPOWER SC

- Supported stability of the **Dimchan All-in-One** POS/ERP ecosystem across F&B/retail businesses; maintained over 95% on-site resolution.
- Executed end-to-end manual testing on web and mobile releases; documented structured defect tickets with reproduction logs to accelerate bug triage.

08/2025 – 10/2025

IT Support & System Deployment Specialist (Project Contract)

HCMC Digital Transformation Center (DXCENTER)

- **Business Process Training (First 2 Weeks):** Recruited for web technical expertise under a municipal digital project; mastered end-to-end workflows of 2 public portals (motcua.tphcm.gov.vn, qlvb.tphcm.gov.vn).
- **Field Stationed at Binh Khanh Ward (1.5 Months):** Directly trained civil servants and citizens on both portals; handled user inquiries, reproduced edge-case bugs, and dispatched structured bug reports to L2 Developers.
- **Hardware, Networking & Digital Signatures:** Troubleshot PC hardware, office LAN networks, and configured digital authentication/signatures; fulfilled deployment contract milestones.